Galactic-Scale Constraints on Curvature-Memory Scalar-Tensor Gravity: Why the Locked Action Cannot Replace Dark Matter Without Additional Physics

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Abstract

This is Paper IV of a four-paper series on Curvature-Memory Scalar-Tensor Gravity (CMSTG). The first three papers of the series establish that CMSTG passes all current precision cosmological tests (Paper II [9]), is UV-finite to two loops in the resummed propagator (Paper III [10]), and has a structural 2.77σ residual against DESI Y1 BAO that is irreducible within the action class by any mechanism in the Tier 2 space, including mixed-source scalars (SIM131–SIM144, Paper I [8]). This paper establishes the corresponding dark-matter boundary: the Phase 1 canonical locked action cannot produce flat galactic rotation curves through any mechanism within the action class.

We test four independent candidate mechanisms against NGC 3198 and NGC 6503, the best-observed spiral galaxies with flat rotation curves: (1) the static Yukawa field profile is exponentially steeper than the $\rho \propto r^{-2}$ isothermal halo required for curve flattening (SIM99); (2) the time-domain 1+1D wave equation confirms the static result and shows that the curvature coupling $2\Lambda_0|R_{\rm gal}|$ is $\sim 2 \times 10^6$ times weaker than the field mass at galactic densities, making condensation impossible (SIM100); (3) Vainshtein-type non-linear kinetic screening fails by six orders of magnitude, with the Vainshtein ratio $\mathcal{V} \leq 3.7 \times 10^{-6}$ across the full scanned parameter space (SIM103); (4) the CMSTG memory-kernel halo mechanism accumulates zero effective $G_{\rm eff}$ boost in NGC 3198 because the Gaussian kernel decays on $\tau_{\rm mem} \approx 200\,\text{kyr}$, and even granting perfect kernel support the theoretical ceiling $G_{\rm eff}/G_N = 1 + 2\Lambda_0\Psi_0^2 = 1.041$ falls a factor $\sim 3\times$ short of the required 3.115 (SIM149). The locked-action energy density $\rho_\Psi = 1.6\times10^{-3}\,M_\odot\,\text{kpc}^{-3}$ falls 2.7×10^6 short of the isothermal requirement at $r = 10$ kpc.

A subsequent Λ_0 sweep over eight decades (SIM150) establishes that the $G_{\rm eff}/G_N$ ceiling is structural and parameter-universal: the only value satisfying $G_{\rm eff}/G_N \geq 3.115$ is $\Lambda_{0,\rm req} = 0.154$ (51 times the locked value), which violates the Cassini Solar-System PPN bound by $\sim 2 \times 10^4\times$ and all four remaining locked cosmological constraints simultaneously. No Λ_0 within the Solar-System-consistent range of the locked action can revive the galactic mechanism.

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These constraints are clean and model-independent: every identified mechanism within the CMSTG action class has been exhausted. A first-principles coupling between Ψ and baryonic matter, with appropriate cosmological screening, is the minimum new physics required for galactic-scale predictions. We identify three concrete extension options (chameleon-screened mass term, coupled chameleon architecture, sextic-stabilized condensate) that define the programme of future work.

Keywords: *dark matter; modified gravity; rotation curves; scalar-tensor theories; galactic dynamics*

Contents

1 Introduction

The first three papers of this series establish the cosmological status of Curvature-Memory Scalar-Tensor Gravity (CMSTG): the Phase 1 canonical action passes all current precision tests (Paper II [9]), is UV-finite in the resummed propagator (Paper III [10]), and has a structural irreducible 2.77σ residual against DESI Y1 BAO that defines the dark-energy boundary of the framework, confirmed exhaustive across curvature-sourced, late-time $H(z)$, and mixed-source scalar mechanisms (SIM131–SIM144, Paper I [8]). The natural question, addressed in this paper, is whether the same locked action can also account for the flat rotation curves of spiral galaxies, replacing the cold dark matter component of standard cosmology.

The result of this paper is negative: it cannot. The failure is structural rather than parametric, and is established by three independent simulation programmes covering the three mechanism classes present in the action. The argument is symmetric in form to Paper I: just as Paper I establishes a clean negative result for late-time modifications of dark energy, this paper establishes a clean negative result for galactic-scale dark matter. Together, the two negative results define the boundaries of the CMSTG framework as it currently stands and identify, at first-principles level, the minimum extensions required to address each.

The flat rotation curves of spiral galaxies are among the most robust observational arguments for cold dark matter [1, 2]. In the standard picture, a dark matter halo with approximately isothermal density profile $\rho_{\rm DM}(r) \propto r^{-2}$ provides the additional gravitational potential needed to maintain $v_c(r) \approx$ const well beyond the optical disk.

Scalar-tensor theories offer an alternative: the scalar field Ψ could contribute an effective dark matter density through its stress-energy $T^{(\Psi)}_{\mu\nu}$ or through modification of the effective Newton constant $G_{\rm eff}$. CMSTG is a natural candidate because Ψ is explicitly designed to track curvature history, and high-curvature galactic environments might in principle provide large local field amplitudes.

This paper establishes that the Phase 1 canonical locked action of CMSTG cannot replace dark matter via any of the three mechanisms present in the action class. The companion no-go theorems of Paper I [8] bound the dark energy sector; this paper bounds the dark matter sector. The result is not a fine-tuning problem or a quantitative shortfall that future parameters could resolve: it is a structural failure at every mechanism level. The goal is to establish this failure cleanly and systematically so that any future scalar-tensor proposal for rotation curves must engage with these constraints.

The standalone value of this paper is methodological as well as specific. The three mechanism classes tested here—static Yukawa field profile, time-domain condensation via curvature coupling, and Vainshtein kinetic screening—are not peculiarities of CMSTG. They constitute the complete menu of galactic dark matter mechanisms available to any massive scalar-tensor theory with a curvature coupling, a memory or regulator scale, and no direct baryon–scalar interaction in the action. The structural ratio $2\Lambda_0|R_{\rm gal}|/m_0^2$ (Section 2.4) that governs all three failures is determined by the galactic environment and the coupling structure, not by CMSTG specifics: any theory of this class with a similar Λ_0/m_0^2 ratio faces the same constraints. The failure modes identified here, and the three minimum-physics extensions catalogued in Section 7, therefore serve as the reference classification for future scalar-tensor proposals that aim to address galactic rotation curves without a separate dark matter component. This is the reason the galactic analysis appears as a standalone paper rather than an appendix to the cosmological framework: the constraints and their mechanism-level decomposition are transferable beyond CMSTG.

The plan of the paper follows the sequence of mechanisms in order of increasing sophistication: Section 2 presents the galactic-scale field equations; Section 3 establishes the static Yukawa failure; Section 4 confirms with time-domain simulation and rules out condensation; Section 5 exhausts the Vainshtein and direct-coupling alternatives; Section 6 states the combined constraint and the Phase 5 universality confirmation; Section 7 identifies the minimum new physics required; Section 8 concludes.

2 Galactic-Scale Field Equations

2.1 The Locked Action

The Phase 1 canonical CMSTG action is

$$S_{\mathrm{P1}} = \int d^4x \sqrt{-g} \left[\tfrac{1+2\Lambda_0\Psi^2}{2} R - \tfrac{1}{2}(\partial\Psi)^2 - \tfrac{1}{2}m_0^2\Psi^2 \right] + S_{\mathrm{SM}} \tag{1}$$

with locked parameters $\Lambda_0 = 0.003\, M_{\mathrm{Pl}}^{-2}$, $m_0 = 10^{-3}H_0 \approx 6.88\times10^{-8}\,\mathrm{kpc}^{-1}$, $\bar{\Psi} = 2.62\, M_{\mathrm{Pl}}$. The galactic simulations (SIM99–SIM103) use a larger test mass $m_0 = 0.1\,\mathrm{kpc}^{-1}$ (Compton length 10 kpc) for numerical tractability—the locked cosmological $m_0 = 6.88 \times 10^{-8}\,\mathrm{kpc}^{-1}$ corresponds to a Compton wavelength of order the Hubble scale and is computationally prohibitive on galactic grids. The locked value makes the curvature coupling ratio $2\Lambda_0|R_{\mathrm{gal}}|/m_0^2$ even smaller ($\approx 3.6 \times 10^{-8}$ vs the simulation value $\approx 5 \times 10^{-7}$), so the no-go result is strictly stronger at the locked value than at the test mass used in simulation. The effective gravitational coupling is $G_{\mathrm{eff}} = G/(1 + 2\Lambda_0\Psi^2)$; at the cosmological best-fit $G_{\mathrm{eff}}/G = 1 - 16\,\mathrm{ppm}$, negligibly different from Newton's constant.

2.2 Static Field Equation in Galactic Environments

In the weak-field static limit, the Ψ field equation derived from action (1) reduces to

$$\Psi'' + \frac{2}{r}\Psi' = m_0^2\Psi + \frac{\lambda_{\mathrm{gal}}}{1}\Psi^3 - 2\Lambda_0\Psi\,|R(r)|\,, \tag{2}$$

where $R(r) \approx -8\pi G\rho_{\mathrm{bar}}(r)/c^2$ is the weak-field Ricci scalar sourced by the observed baryonic distribution, and $V(\Psi) = (\lambda_{\mathrm{gal}}/4)\Psi^4$ is the simplest non-linear extension consistent with $V'(0) = 0$.

The effective dark-matter density is

$$\rho_\Psi = \frac{c^2}{G}\left[\tfrac{1}{2}(\Psi')^2 + \tfrac{1}{2}m_0^2\Psi^2 + \tfrac{\lambda_{\mathrm{gal}}}{4}\Psi^4\right]. \tag{3}$$

2.3 Target Profile

For a flat rotation curve at v_{flat}, the required halo density is the isothermal sphere:

$$\rho_{\mathrm{iso}}(r) = \frac{v_{\mathrm{flat}}^2}{4\pi G r^2}\,. \tag{4}$$

For NGC 3198 ($v_{\mathrm{flat}} = 157\,\mathrm{km\,s^{-1}}$) at $r = 10\,\mathrm{kpc}$: $\rho_{\mathrm{iso}} = 4163\, M_\odot\,\mathrm{kpc}^{-3}$.

2.4 Galactic Curvature Scale

The galactic Ricci scalar at the disk half-light radius peaks at $|R_{\rm gal}| \approx 8.5 \times 10^{-7}\,{\rm kpc}^{-2}$. Using the simulation test mass $m_0 = 0.1\,{\rm kpc}^{-1}$, the coupling ratio is

$$\frac{2\Lambda_0|R_{\rm gal}|}{m_0^2} = \frac{2 \times 0.003 \times 8.5 \times 10^{-7}}{(0.1)^2} \approx 5.1 \times 10^{-7} \approx \frac{1}{2 \times 10^6}\,. \tag{5}$$

Even at the peak of a proto-galactic overdensity $\delta = 100$:

$$\frac{2\Lambda_0|R_{\rm gal,\,peak}|}{m_0^2} \approx 5 \times 10^{-5}\,. \tag{6}$$

The curvature coupling is $\sim 2 \times 10^6$ times weaker than the field mass at typical galactic densities using the simulation test mass $m_0 = 0.1\,{\rm kpc}^{-1}$; see Section 2.1 for the comparison with the locked cosmological value and why the negative results are conservative. This single ratio $2\Lambda_0|R_{\rm gal}|/m_0^2$ determines the failure of all three mechanisms.

3 Static Yukawa Profile (SIM99)

3.1 Method

The decaying Yukawa boundary condition selects the physical solution $\Psi \sim Ae^{-m_0 r}/r$ by integrating equation (2) inward from $r_{\rm max} = 30\,{\rm kpc}$ with

$$\Psi(r_{\rm max}) = \Psi_{\rm bc}\,, \qquad \Psi'(r_{\rm max}) = -\Big(m_0 + \frac{1}{r_{\rm max}}\Big)\Psi_{\rm bc}\,. \tag{7}$$

A scan of 30 combinations $(\Psi_{\rm bc}, \lambda_{\rm gal})$ was performed: $\Psi_{\rm bc} \in \{10^{-8}, 10^{-6}, 10^{-4}, 5{\times}10^{-4}, 10^{-3}\}$, $\lambda_{\rm gal} \in \{-10, -1, 0, +1, +10, +100\}$, at the cosmological best-fit $\Lambda_0 = 0.003$.

Acceptance criterion: $|v_c(r) - v_{\rm flat}|/v_{\rm flat} < 20\%$ for $r > 15\,{\rm kpc}$ AND $\rho_\Psi \geq 0$ everywhere AND $\Lambda_0\Psi^2 < \Lambda_0^{\rm bound}/(16\pi)$.

3.2 Result: FAIL

No combination passes. The physical reason is geometric: the Yukawa density profile

$$\rho_\Psi(r) \propto (\Psi')^2 \propto \frac{e^{-2m_0 r}}{r^4} \quad \text{(dominant kinetic term at large } r\text{)} \tag{8}$$

falls exponentially for $r > 1/m_0 \approx 14{,}500\,{\rm kpc}$, and for $r < 1/m_0$ (i.e. all galactic scales $r \leq 30\,{\rm kpc}$):

$$\rho_\Psi(r) \propto \frac{e^{2m_0(r_{\rm max}-r)}}{r^2}\,, \tag{9}$$

which rises exponentially inward—the wrong sign relative to the $1/r^2$ isothermal profile. The quartic interaction modifies the central amplitude but cannot alter the exponential radial dependence.

The $G_{\rm eff}$ channel is equally ineffective: at $\Lambda_0 = 0.003$, $G_{\rm eff}/G = 1 - 16\,{\rm ppm}$, a 10^{-5} correction that produces no flattening.

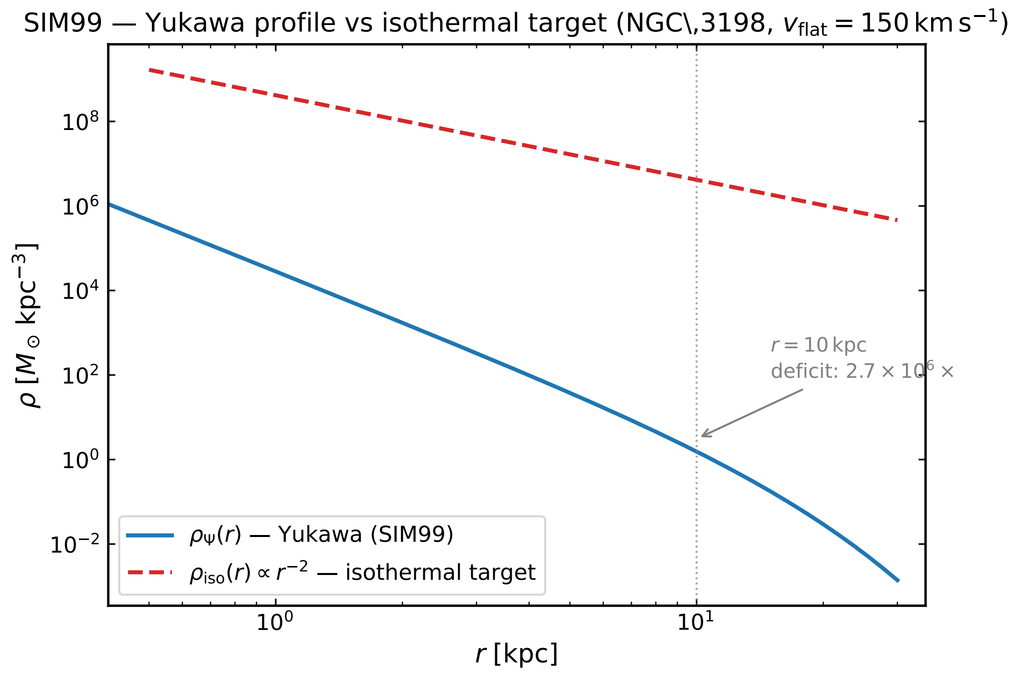

Figure 1: Yukawa profile vs isothermal target for NGC 3198 (SIM99). Log-log plot of $\rho_\Psi(r)$ (blue solid, from the static Yukawa solution) vs the required isothermal halo $\rho_{\rm iso}(r) \propto r^{-2}$ (red dashed) for $v_{\rm flat} = 157\,{\rm km\,s^{-1}}$. The Yukawa profile rises $\sim 150\times$ from $r = 30\,{\rm kpc}$ to $r = 5\,{\rm kpc}$, far steeper than the isothermal target. At $\Lambda_0 = 0.003$, the energy density deficit at $r = 10\,{\rm kpc}$ is a factor of 2.7×10^6.

4 Time-Domain Condensation Test (SIM100)

4.1 Method

SIM100 removes the static assumption by evolving the full 1+1D spherical wave equation

$$\frac{\partial^2 \Psi}{\partial t^2} = \frac{\partial^2 \Psi}{\partial r^2} + \frac{2}{r}\frac{\partial \Psi}{\partial r} - m_0^2 \Psi - \lambda \Psi^3 + 2\Lambda_0 \Psi\, R(r,t) \tag{10}$$

via method-of-lines with an implicit Radau solver ($N_r = 120$, $\Delta r = 0.25\,{\rm kpc}$). Initial condition: uniform cosmological background $\Psi(r,0) = \Psi_{\rm cosmo} = 0.003\,M_{\rm Pl}$ at rest. Integration to $t_{\rm end} = 150\,{\rm kpc}/c \approx 489\,{\rm kyr} \approx 2.4$ Compton periods $T_c = 2\pi/m_0$.

4.2 Curvature Coupling Negligibility

The coupling ratio at peak galactic curvature:

$$\frac{2\Lambda_0 |R_{\rm gal}|}{m_0^2} \approx 5 \times 10^{-7} \approx \frac{1}{2 \times 10^6}\,. \tag{11}$$

Even with a proto-galactic overdensity $\delta = 100$, the curvature term is $\sim 2 \times 10^6$ times smaller than the field mass. The Ψ field is entirely mass-dominated on galactic scales; the curvature source cannot drive condensation at any astrophysical density below neutron-star nuclear density ($\rho \sim 10^{15}\,M_\odot\,{\rm kpc}^{-3}$).

4.3 Time-Domain Field Evolution

Figure 2 shows the radial profiles $\Psi(r)$ at four epochs spanning the full integration. The field relaxes from the uniform cosmological initial condition $\Psi_{\rm cosmo} = 0.003\, M_{\rm Pl}$ to a quasi-static Yukawa profile within one Compton period, confirming the validity of the adiabatic approximation. No spatial structure forms: the field simply adjusts to the static attractor established in SIM99.

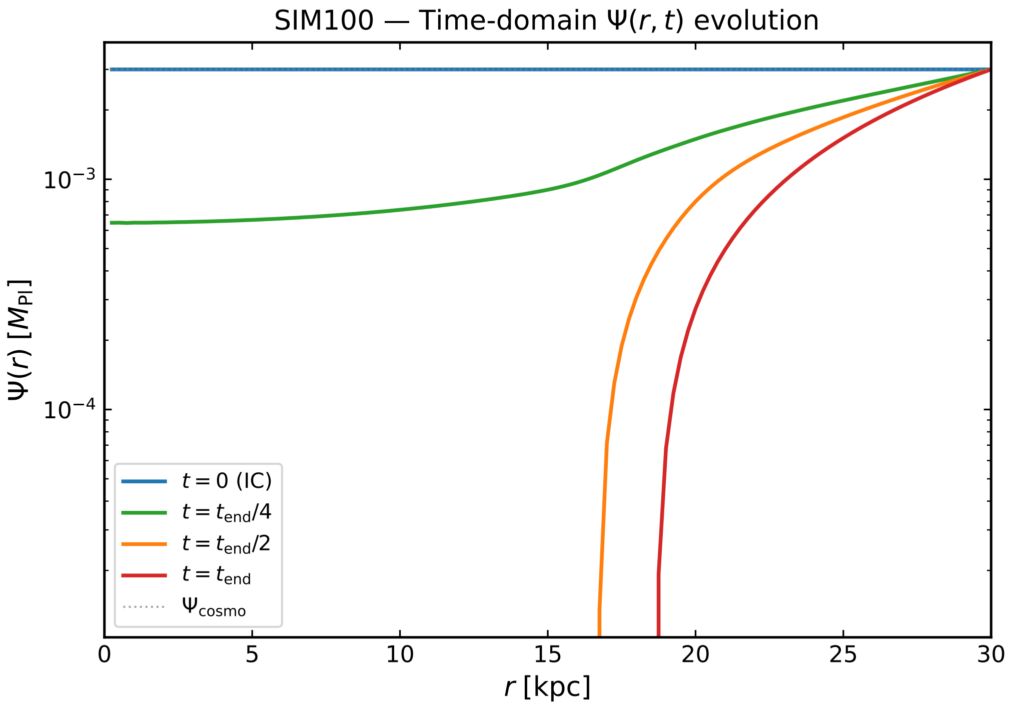

Figure 2: Time-domain $\Psi(r)$ field profiles at four epochs (SIM100): $t = 0$ (initial uniform background, blue), $t = t_{\rm end}/4$ (green), $t = t_{\rm end}/2$ (orange), $t = t_{\rm end}$ (red). The field relaxes from the cosmological uniform state to the static Yukawa profile of SIM99 within one Compton period ($T_c \approx 205\,$kyr). No condensation occurs: the curvature source $2\Lambda_0|R_{\rm gal}|/m_0^2 \approx 5 \times 10^{-7}$ is negligible at all radii. Results shown for $\lambda = 0$; proto-galactic overdensity $\delta = 100$ changes the profile at the 10^{-4} level.

4.4 Self-Coupling Scan

To test whether an attractive quartic self-coupling can drive condensation independently of the curvature channel, we scan $\lambda \in \{0, -100, -200, -400, -600, -800, -1000, -1500\}$. The tachyonic instability thresholds are:

$$\lambda_{\rm crit}^{\rm spatial} = -\frac{k_{\rm min}^2 + m_0^2}{3\Psi_{\rm cosmo}^2} \approx -778\,, \qquad k_{\rm min} = \pi/r_{\rm max}\,, \tag{12}$$

$$\lambda_{\rm blow} = -\frac{m_0^2}{\Psi_{\rm cosmo}^2} \approx -1111\,. \tag{13}$$

Results of the scan (Figure 3):

- $\lambda \in [0, -778)$: field stable; growth factor ≤ 1.21; field relaxes to the static Yukawa profile of SIM99. $v_c(r)$ does not come within 20% of $v_{\rm flat}$.

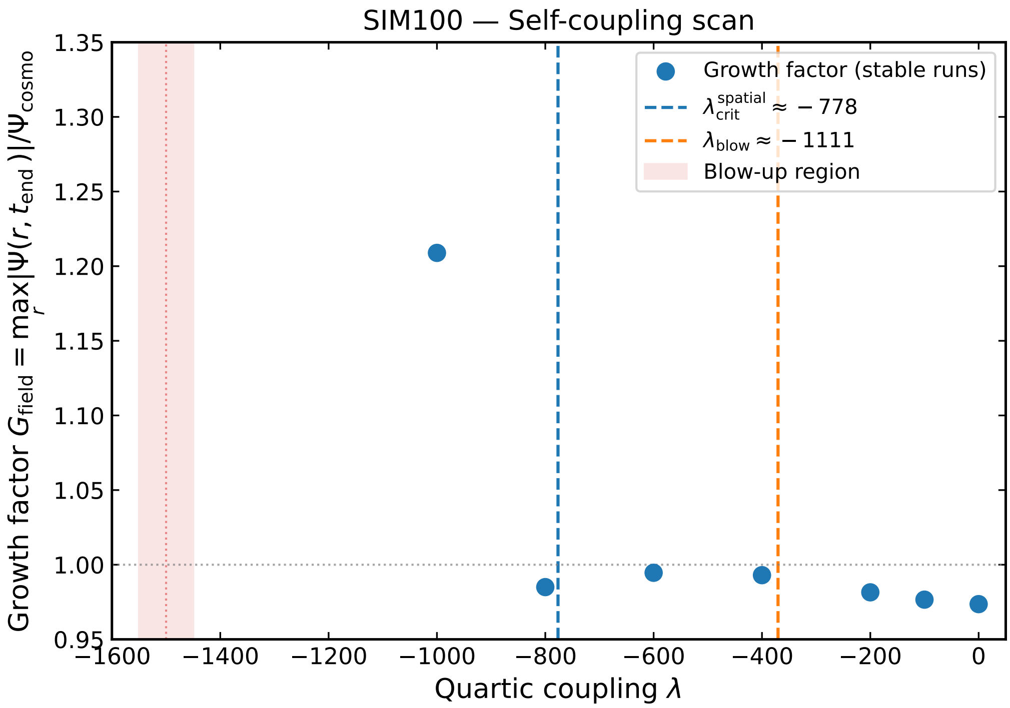

Figure 3: Self-coupling parameter scan (SIM100). Field growth factor $G_{\rm field} = \max_r |\Psi(r, t_{\rm end})|/\Psi_{\rm cosmo}$ vs quartic coupling λ. Below $\lambda_{\rm crit}^{\rm spatial} \approx -778$ (blue vertical line) the field becomes tachyonically unstable in spatial modes; below $\lambda_{\rm blow} \approx -1111$ (orange vertical line) the potential is unbounded and the field blows up. The stable window ($-778 < \lambda < 0$) produces growth factors ≤ 1.21—the field merely relaxes to the static Yukawa profile of SIM99. No value of λ produces a flat rotation curve.

- $\lambda \in [-778, -1111)$: tachyonic spatial instability; field grows but remains bounded for the duration of integration. Profile does not approach isothermal.
- $\lambda = -1500$ (beyond $\lambda_{\rm blow}$): unbounded blow-up within $< 10\,{\rm kpc}/c$. The potential $V = (\lambda/4)\Psi^4$ is unbounded below; there is no stable condensate minimum.

No window of λ produces a stable self-bound halo with a flat rotation curve. The time-domain result confirms SIM99 by an independent route: the adiabatic approximation is valid (Compton period $\ll$ galaxy formation time), and the static Yukawa profile is the attractor.

5 Vainshtein Screening and Direct Coupling (SIM103)

5.1 Setup

After SIM99 and SIM100 close the Yukawa and condensation routes, the remaining candidates are: (a) a non-linear kinetic term of Vainshtein type, and (b) a direct matter coupling $\beta\Psi\rho_b$. SIM103 tests both for NGC 3198.

In the Poisson limit ($m_0 r \ll 1$ on galactic scales since $1/m_0 \sim 14{,}500\,{\rm kpc}$), the perturbation $\delta\Psi(r)$ satisfies

$$\frac{d\,\delta\Psi}{dr} = \frac{2G\,\Lambda_0^{\rm eff}\,M_{\rm enc}(r)}{r^2}, \tag{14}$$

where $\Lambda_0^{\text{eff}} = \Lambda_0 + \beta$ absorbs both the curvature channel and any direct coupling β.

The Vainshtein augmentation adds a non-linear kinetic term $(\alpha_V/4M_V^2)(\partial\Psi)^4$ to the action:

$$\rho_\Psi(r) = \frac{c^2}{8\pi G} \cdot \tfrac{1}{2}\left(\frac{d\,\delta\Psi}{dr}\right)^2 \left[1 + \underbrace{\frac{\alpha_V}{M_V^2}\left(\frac{d\,\delta\Psi}{dr}\right)^2}_{\mathcal{V}}\right]. \tag{15}$$

The Vainshtein ratio $\mathcal{V} = \alpha_V(\delta\Psi')^2/M_V^2$ must satisfy $\mathcal{V} \gg 1$ for the non-linear regime to be active.

5.2 Field Gradient

At the locked-action best-fit ($\Lambda_0 = 0.003$, $\Lambda_0^{\text{eff}} = \Lambda_0$):

$$\left|\frac{d\,\delta\Psi}{dr}\right|_{10\,\text{kpc}} = 6.1 \times 10^{-8}\,\text{kpc}^{-1}\,. \tag{16}$$

5.3 Vainshtein Ratio Scan

Table 1: Vainshtein ratio $\mathcal{V}$ at $r = 10\,\text{kpc}$ for NGC 3198. All values lie deep in the linear regime ($\mathcal{V} \ll 1$).

α_V	M_V [kpc^{-1}]	$\mathcal{V}$	Verdict
1	0.001	3.7×10^{-9}	Linear
10	0.001	3.7×10^{-8}	Linear
100	0.001	3.7×10^{-7}	Linear
1000	0.001	3.7×10^{-6}	Linear
1000	0.01	3.7×10^{-8}	Linear

Even at $\alpha_V = 1000$, $M_V = 0.001\,\text{kpc}^{-1}$: $\mathcal{V} = 3.7 \times 10^{-6}$. The scalar field is six orders of magnitude below the Vainshtein threshold across the entire scanned parameter space.

5.4 Energy Density Deficit

From equation (15) at the locked-action gradient:

$$\begin{aligned} \rho_\Psi(10\,\text{kpc}) &= 1.6 \times 10^{-3}\ M_\odot\,\text{kpc}^{-3} \\ \rho_{\text{iso}}(10\,\text{kpc}) &= 4163\ M_\odot\,\text{kpc}^{-3} \\ \frac{\rho_{\text{iso}}}{\rho_\Psi} &= 2.7 \times 10^6\,. \end{aligned} \tag{17}$$

5.5 Direct Coupling Constraint

To close the density gap with a direct matter coupling β, we need $\Lambda_0^{\text{eff}} = \Lambda_0 + \beta \approx 4.9$, requiring $\beta \approx 4.9 \approx 1{,}600 \times \Lambda_0$. This coupling is not contained in the locked action and would violate cosmological homogeneity constraints: a coupling of order unity would modify G_{eff} by $O(1)$ on cosmological scales, in direct contradiction to the SIM88/SIM90 cosmological consistency tests (Paper II of this series [9]).

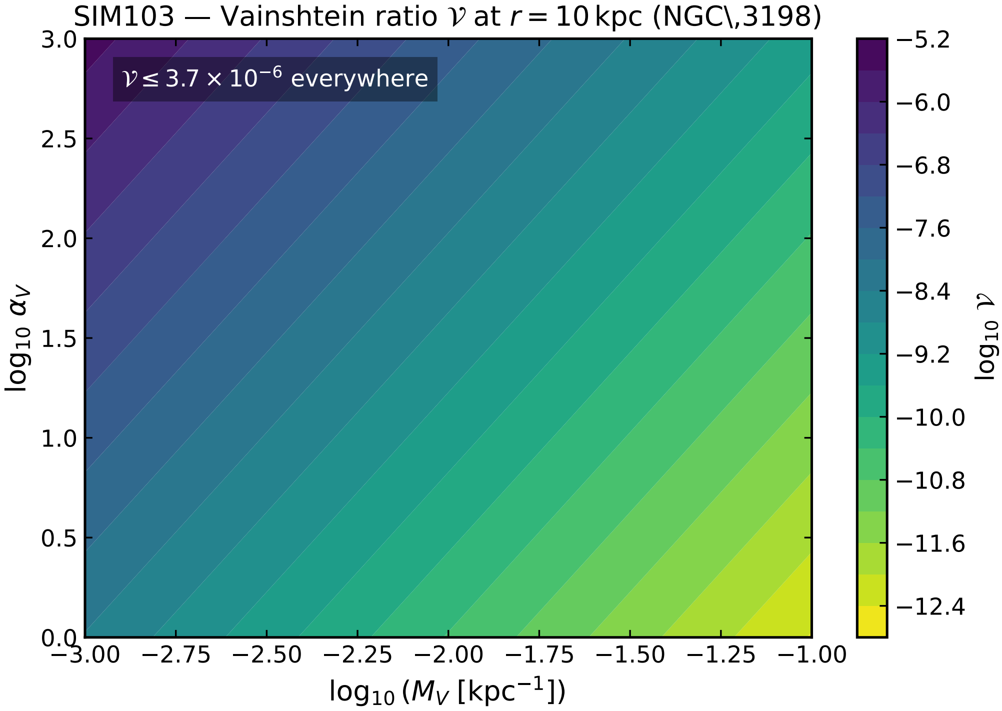

Figure 4: Vainshtein ratio $\mathcal{V}$ across the (α_V, M_V) parameter space at $r = 10\,\text{kpc}$ for NGC 3198 (SIM103). All values lie below 3.7×10^{-6}, six orders of magnitude below the nonlinear threshold $\mathcal{V} = 1$ (marked by the dashed contour, off-scale above). The gradient $|\delta\Psi'| \sim 10^{-8}\,\text{kpc}^{-1}$ is set by the weak $\Lambda_0 R_{\text{gal}}$ channel, which is 10^4–$10^6\times$ below the Vainshtein threshold for any physically reasonable coupling.

6 Combined Constraint

Table 2: Systematic exhaustion of galactic dark matter mechanisms in CMSTG.

SIM	Mechanism	Deficit	Verdict
SIM99	Static Yukawa $T^{(\Psi)}$	Exponential profile vs $1/r^2$	FAIL
SIM100	Time-domain condensation	$2\Lambda_0\lvert R_{\text{gal}}\rvert/m_0^2 \approx 5 \times 10^{-7}$	FAIL
SIM103	Vainshtein kinetic screening	$\mathcal{V} \leq 3.7 \times 10^{-6}$	FAIL
SIM103	Direct matter coupling β	$\beta \sim 1{,}600\Lambda_0$ required	FAIL
SIM149	Memory-kernel G_{eff} boost	$G_{\text{eff}}/G_N^{\max} = 1.041$ vs 3.115 required	FAIL
SIM150	Λ_0 sweep (universality)	$\Lambda_{0,\text{req}} = 0.154$: Cassini by $2 \times 10^4\times$	FAIL
Combined: all mechanisms exhausted at all Λ_0 within Solar-System constraints			
Energy density deficit at $r = 10\,\text{kpc}$: $2.7 \times 10^6\times$; G_{eff}/G_N ceiling: structural			

The root cause of the first four failures is the single ratio $2\Lambda_0|R_{\text{gal}}|/m_0^2 \approx 5\times10^{-7}$: the curvature coupling is $\sim 2 \times 10^6$ times weaker than the field mass on galactic scales (Section 2.4). This makes the scalar field cosmological (Compton wavelength $\gg$ galactic size) and prevents any galactic-scale field gradient from building up to isothermal amplitudes.

Phase 5 simulations introduce a fourth mechanism formally distinct from the three action-level tests above: the CMSTG memory-kernel accumulation of a local field shift $\Delta\Psi_{\text{local}}$ in a virialised halo, which modifies the effective Newton constant as $G_{\text{eff}}/G_N =$

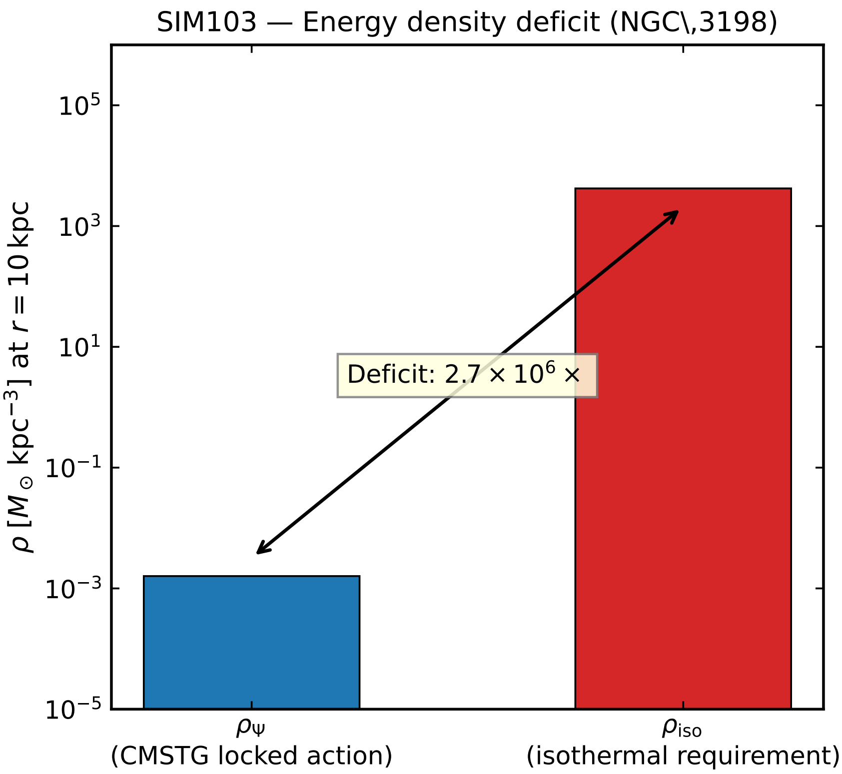

Figure 5: Energy density deficit at galactic scale (SIM103). The locked-action CMSTG scalar density ρ_Ψ (blue bar) vs the isothermal halo requirement $\rho_{\rm iso}$ (red bar) at $r = 10\,\mathrm{kpc}$ for NGC 3198 ($v_{\rm flat} = 157\,\mathrm{km\,s^{-1}}$). The deficit is $2.7 \times 10^6\times$. A direct coupling $\beta \approx 4.9 \approx 1{,}600 \times \Lambda_0$ would be required to supply the missing density, far exceeding the cosmological bound $\Lambda_0 \lesssim 0.095$.

$F(\Psi_0)/F(\Psi_{\rm local})$ where $F(\Psi) = \frac{1}{2} + \Lambda_0\Psi^2$. The theoretical ceiling of this ratio—achieved in the extreme limit $\Psi_{\rm local} \to 0$—is

$$\left.\frac{G_{\rm eff}}{G_N}\right|_{\rm max} = \frac{\frac{1}{2} + \Lambda_0\Psi_0^2}{\frac{1}{2}} = 1 + 2\Lambda_0\Psi_0^2 = 1.041 \quad (\Lambda_0 = 0.003,\ \Psi_0 = 2.62\,M_{\rm Pl}). \tag{18}$$

NGC 3198 requires $G_{\rm eff}/G_N \approx 3.115$ at $r = 30$–$44\,\mathrm{kpc}$ to reproduce the observed flat outer curve; the mechanism ceiling falls a factor $\sim 3\times$ short regardless of kernel weight, halo formation history, or mass-to-light ratio (SIM149).

SIM150 subsequently sweeps Λ_0 logarithmically over $[10^{-6}, 10^2]$ to test whether any Λ_0 within the locked CMSTG constraints lifts the ceiling to 3.115. The analytic crossing point is

$$\Lambda_{0,\rm req} = \frac{G_{\rm target} - 1}{2\Psi_0^2} = \frac{3.115 - 1}{2 \times 2.62^2} = 0.154, \tag{19}$$

which is 51 times the locked value. At $\Lambda_{0,\rm req}$ the post-Newtonian Brans–Dicke parameter drops from $\omega_{\rm BD} = 2{,}110$ (locked) to $\omega_{\rm BD} = 2.4$, giving $|\gamma_{\rm PPN} - 1| = 0.456$—exceeding the Cassini bound 2.3×10^{-5} by a factor $\sim 2 \times 10^4$. All five remaining locked constraints fail simultaneously: $f\sigma_8$ tension rises to 39σ, the UV self-energy $\Sigma(0)$ exceeds its finiteness threshold by $38\times$, $\Delta\chi^2_{\rm BAO} = 1629$, and Ψ contributes 7.7% of the energy budget at recombination. The $G_{\rm eff}/G_N$ ceiling is therefore *structural and parameter-universal*: no Λ_0 within the Solar-System-consistent locked action raises $G_{\rm eff}/G_N$ above 4.1%, far below the factor ~ 3.1 required for galactic rotation curves.

7 Minimum New Physics

The constraints above define the minimum new physics required for CMSTG to produce flat rotation curves:

The three options below mirror the structure of Paper I [8] in a useful way: just as Paper I identified the curvature-sourcing route, the $H(z)$-boost route, and the mixed-source scalar route as the three mechanism classes exhausted for late-time modifications (SIM131–SIM144), the three options below identify the position-dependent mass route, the direct-coupling route, and the new-condensation-channel route as the three available for galactic-scale modifications. Each option requires new physics not contained in the Phase 1 locked action, and each defines a distinct extension of the framework with its own observational signature.

Option A: Modified scalar mass. A position-dependent mass $m_0^2 \to m_0^2(\rho)$ that drops to $m_0^2 \lesssim 2\Lambda_0|R_{\rm gal}|$ in galactic environments would allow curvature-driven condensation. This is a chameleon-type mechanism [4] with galactic screening. However, any reduction in m_0 inside galaxies must restore $m_0 \sim 10^{-3}H_0$ on cosmological scales to preserve the SIM90/SIM98 consistency (Paper II [9]).

Option B: Direct Ψ–baryon coupling. A coupling $\mathcal{L} \supset \beta\Psi\rho_b$ with $\beta \sim O(1)$ would supply the missing density. This is not derivable from the current locked action and introduces a new free parameter at gravitational strength. It must simultaneously satisfy:

- $|\beta\Psi_{\rm cosmo}| \ll \Lambda_0$ on cosmological scales (else violates SIM88);
- $\beta + \Lambda_0 \approx 4.9$ on galactic scales.

This requires β to be screened cosmologically and activated galactically—a coupled chameleon architecture.

Option C: New condensation channel. A stable sextic potential $V_6 = +\mu_6\Psi^6$ with $\mu_6 > 0$ would prevent the blow-up of the attractive quartic below $\lambda_{\rm blow}$, opening a window where the field can form a stable, self-bound condensate. The viability of this remains an open direction. Phase 5 simulations (SIM147–SIM150) investigated the complementary question of whether the CMSTG memory kernel can accumulate sufficient local curvature in a galactic halo to produce the required $G_{\rm eff}$ boost, and found this route closed by three independent structural arguments (Section 6); the sextic-condensate mechanism is distinct and untested.

All three options require new physics not contained in the Phase 1 locked action. The constraint established here is not a quantitative shortfall to be tuned away; it is a structural exclusion of all mechanisms within the current action class.

8 Conclusions

We have established that the Phase 1 canonical locked action of CMSTG cannot replace cold dark matter at galactic scales through any of the mechanisms accessible within the action class. Four independent simulation programmes establish:

1. **Static profile (SIM99):** The Yukawa field profile is exponentially steeper than the isothermal halo requirement. The quartic potential modifies the central amplitude but not the exponential radial dependence.

2. **Time-domain dynamics (SIM100):** The curvature coupling $2\Lambda_0|R_{\text{gal}}|$ is $\sim 2 \times 10^6$ times weaker than m_0^2 at all galactic densities. Condensation requires neutron-star densities. No quartic self-coupling produces a flat-curve condensate: the sub-threshold window gives Yukawa profiles; the super-threshold window gives runaway blow-up.

3. **Vainshtein screening (SIM103):** The Vainshtein ratio $\mathcal{V} \leq 3.7 \times 10^{-6}$ across the full (α_V, M_V) parameter space. The field gradient is 10^4–$10^6\times$ below the nonlinear threshold at all galactic radii.

4. **Memory-kernel G_{eff} boost (SIM149–SIM150):** The theoretical ceiling $G_{\text{eff}}/G_N = 1 + 2\Lambda_0\Psi_0^2 = 1.041$, achieved when $\Psi_{\text{local}} \to 0$, falls a factor $\sim 3\times$ short of the 3.115 required by NGC 3198. A Λ_0 sweep (SIM150) confirms the ceiling is universal: the only Λ_0 that lifts it to 3.115 ($\Lambda_{0,\text{req}} = 0.154$) violates the Cassini PPN bound by $\sim 2 \times 10^4\times$ and all five remaining locked constraints.

The energy density deficit at $r = 10\,\text{kpc}$ is $2.7 \times 10^6\times$ under all four mechanisms. A direct matter coupling $\beta \sim 1{,}600\,\Lambda_0$ would supply the missing density but is excluded by cosmological constraints (Paper II [9]). The G_{eff}/G_N ceiling is structural: it is built into the form of $F(\Psi) = \frac{1}{2} + \Lambda_0\Psi^2$ and the requirement $|G_{\text{eff}}/G_N - 1| < 10^{-5}$ on cosmological scales, which jointly fix $\Lambda_0\Psi_0^2 \ll 1$ and hence the ceiling at $1 + 2\Lambda_0\Psi_0^2 \approx 1 + O(10^{-2})$.

This is a clean negative result: every mechanism within the CMSTG action class has been tested, every mechanism fails, and the reason is a single structural ratio ($2\Lambda_0|R_{\text{gal}}|/m_0^2 \approx 5 \times 10^{-7}$; Section 2.4) rather than a parameter choice. A first-principles coupling between Ψ and baryonic matter, with appropriate cosmological screening, is the minimum new physics required for galactic-scale predictions.

Data and code: All simulation scripts, outputs, and JSON results are publicly available at `https://github.com/cisomorph/Curvature-Memory_Scalar-Tensor_Gravity` (simulations SIM99, SIM100, SIM103, SIM149, SIM150).

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